Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
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Ex6

MAN IN THE MIDDLE ATTACK IN DIFFIE HELLMAN KEY EXCHANGE

Code

Diffie_attacker.java

```
import java.io.*;
import java.math.BigInteger;
import java.net.ServerSocket;
import java.net.Socket;
import java.security.SecureRandom;
import java.util.Scanner;

public class diffie_attacker {

    public static void main(String[] args) throws Exception {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter prime number (p): ");
        BigInteger p = new BigInteger(sc.nextLine());

        System.out.print("Enter primitive root (g): ");
        BigInteger g = new BigInteger(sc.nextLine());

        SecureRandom rand = new SecureRandom();

        BigInteger xd1 = new BigInteger(10, rand);
        BigInteger xd2 = new BigInteger(10, rand);

        BigInteger yd1 = g.modPow(xd1, p);
        BigInteger yd2 = g.modPow(xd2, p);

        System.out.println("\nAttacker Private Keys:");
        System.out.println("Xd1: " + xd1);
        System.out.println("Xd2: " + xd2);
```

```java
        System.out.println("\nAttacker Public Keys:");
        System.out.println("Yd1: " + yd1);
        System.out.println("Yd2: " + yd2);

        ServerSocket server = new ServerSocket(6000);
        System.out.println("\nAttacker waiting for Alice...");
        Socket alice = server.accept();

        DataInputStream disA = new DataInputStream(alice.getInputStream());
        DataOutputStream dosA = new DataOutputStream(alice.getOutputStream());

        BigInteger YA = new BigInteger(disA.readUTF());
        System.out.println("\nReceived Alice Public Key: " + YA);

        dosA.writeUTF(yd1.toString());  // send fake key

        BigInteger KA = YA.modPow(xd1, p);
        System.out.println("Common key between Alice and Attacker: " + KA);

        System.out.println("\nAttacker waiting for Bob...");
        Socket bob = server.accept();

        DataInputStream disB = new DataInputStream(bob.getInputStream());
        DataOutputStream dosB = new DataOutputStream(bob.getOutputStream());

        BigInteger YB = new BigInteger(disB.readUTF());
        System.out.println("\nReceived Bob Public Key: " + YB);

        dosB.writeUTF(yd2.toString());  // send fake key

        BigInteger KB = YB.modPow(xd2, p);
        System.out.println("Common key between Bob and Attacker: " + KB);

        alice.close();
        bob.close();
        server.close();
    }
}
```

Diffie_alice.java

```java
import java.io.*;
import java.math.BigInteger;
import java.net.Socket;
import java.security.SecureRandom;
import java.util.Scanner;

public class diffie_alice {

    public static void main(String[] args) throws Exception {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter prime number (p): ");
        BigInteger p = new BigInteger(sc.nextLine());

        System.out.print("Enter primitive root (g): ");
        BigInteger g = new BigInteger(sc.nextLine());

        SecureRandom rand = new SecureRandom();
        BigInteger xa = new BigInteger(10, rand);

        BigInteger YA = g.modPow(xa, p);

        System.out.println("\nAlice Private Key: " + xa);
        System.out.println("Alice Public Key: " + YA);

        Socket socket = new Socket("localhost", 6000);

        DataOutputStream dos = new DataOutputStream(socket.getOutputStream());
        DataInputStream dis = new DataInputStream(socket.getInputStream());

        dos.writeUTF(YA.toString());

        BigInteger fakeKey = new BigInteger(dis.readUTF());

        System.out.println("Public key received (from attacker): " + fakeKey);

        BigInteger KA = fakeKey.modPow(xa, p);

        System.out.println("Common key between Alice and Attacker: " + KA);

        socket.close();
    }
}
```

Diffie_bob.java

```java
import java.io.*;
import java.math.BigInteger;
import java.net.Socket;
import java.security.SecureRandom;
import java.util.Scanner;

public class diffie_bob {

    public static void main(String[] args) throws Exception {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter prime number (p): ");
        BigInteger p = new BigInteger(sc.nextLine());

        System.out.print("Enter primitive root (g): ");
        BigInteger g = new BigInteger(sc.nextLine());

        SecureRandom rand = new SecureRandom();
        BigInteger xb = new BigInteger(10, rand);

        BigInteger YB = g.modPow(xb, p);

        System.out.println("\nBob Private Key: " + xb);
        System.out.println("Bob Public Key: " + YB);

        Socket socket = new Socket("localhost", 6000);

        DataOutputStream dos = new DataOutputStream(socket.getOutputStream());
        DataInputStream dis = new DataInputStream(socket.getInputStream());

        dos.writeUTF(YB.toString());

        BigInteger fakeKey = new BigInteger(dis.readUTF());

        System.out.println("Public key received (from attacker): " + fakeKey);

        BigInteger KB = fakeKey.modPow(xb, p);

        System.out.println("Common key between Bob and Attacker: " + KB);

        socket.close();
    }
}
```

Output [screenshot]

Diffie_attacker.java

```
PS E:\aa vit sem6\z_Crypto-lab> javac diffie_attacker.java
PS E:\aa vit sem6\z_Crypto-lab> java diffie_attacker
Enter prime number (p): 29
Enter primitive root (g): 2

Attacker Private Keys:
Xd1: 160
Xd2: 117

Attacker Public Keys:
Yd1: 23
Yd2: 3

Attacker waiting for Alice...

Received Alice Public Key: 11
Common key between Alice and Attacker: 20

Attacker waiting for Bob...

Received Bob Public Key: 15
Common key between Bob and Attacker: 10
PS E:\aa vit sem6\z_Crypto-lab>
```

Diffie_alice.java

```
PS E:\aa vit sem6\z_Crypto-lab> javac diffie_alice.java
PS E:\aa vit sem6\z_Crypto-lab> java diffie_alice
Enter prime number (p): 29
Enter primitive root (g): 2

Alice Private Key: 109
Alice Public Key: 11
Public key received (from attacker): 23
Common key between Alice and Attacker: 20
PS E:\aa vit sem6\z_Crypto-lab>
```

Diffie_bob.java

```
PS E:\aa vit sem6\z_Crypto-lab> javac diffie_bob.java
PS E:\aa vit sem6\z_Crypto-lab> java diffie_bob
Enter prime number (p): 29
Enter primitive root (g): 2

Bob Private Key: 783
Bob Public Key: 15
Public key received (from attacker): 3
Common key between Bob and Attacker: 10
PS E:\aa vit sem6\z_Crypto-lab>
```